Figure 0: Interactive Dashboard Architecture & Analytical User Guide

Tab 1: Global Macroevolutionary Figures

`[ Fig 0 ]` `[ Fig 1A ]` `[ Fig 1B ]` `[ Fig 2 ]` `[ Fig 3 ]`

The Big-Picture Evolutionary Synthesis across 68 Vertebrates

- Fig 0, 0A–0C: Conceptual framework, syntenic retrieval pipeline, metric anatomy, and codon dynamics.
- Fig 1A & 1B: Curated summary across 33 species (1A) and exhaustive 68-vertebrate macroevolutionary atlas (1B).
- Fig 2 & 3: Mechanistic drivers of GC polarity (Fig 2) and time-calibrated phylogenetic synteny chronogram (Fig 3).

Tab 2: 1-on-1 Head-to-Head Comparison

`[ Linear Tracks ]` `[ Inset Map ]` `[ Diagnostic Suite ]`

Direct Pairwise Syntenic & Structural Contrast Between Any Two Species

- Select Species A & B from searchable autocomplete comboboxes (e.g., Human vs. Spotted Gar).
- Synchronized views: Linear Chromosome tracks, Bivariate with Mini Inset, or 6-Panel Diagnostic suites.
- Instant differential readout: ΔCDS GC%, ΔWhole-Gene GC%, chromosome coordinates, and syntenic context.

Tab 3: Single Species Deep Dive

`[ Interval Bounds ]` `[ Flanking Markers ]` `[ Ideograms ]`

Locus Anatomy & Architectural Metadata for Any Individual Genome

- Comprehensive profile of any individual species in the 68-genome database.
- Exact genomic interval bounds, flanking syntenic markers, and curated gene count.
- High-resolution chromosome ideograms displaying exact physical coordinates and centromere locations.

Tabs 4 & 5: Interval Database & Statistical Contrasts

`[ 1,180 Gene Models ]` `[ Wilcoxon Tests ]` `[ Effect Sizes ]`

Curated Gene Models & Hypothesis Testing Suite

- Tab 4 (Database): Filterable, searchable table of all 1,180 verified globin genes with z-scores and coordinates.
- Tab 5 (Statistics): Two-sided Wilcoxon signed-rank and Mann-Whitney U tests with Hodges-Lehmann effect sizes.

Recommended Exploratory Workflow & Keyboard Shortcuts:

- Step 1 (Tab 1): Inspect macroevolutionary distributions ➔ Step 2 (Tab 2): 1-on-1 syntenic contrasts ➔ Step 3 (Tab 3): Deep dive interval architecture
- **Interactive Lightbox: Click any image to open fullscreen • Shortcuts: [Esc] Close • [0] Fit to screen • [+] / [-] Zoom • [1]–[6] Diagnostic panels**